



12.1.4 Wrap-Up: Cool Down

1. My favorite graphical representation to display

numerical univariate data is

- box plot.
- line plot.
- histogram.
- stem and leaf.

It is my favorite because ... _____

_____.

2. My favorite graphical representation to display

univariate categorical data is a

- bar graph.
- circle graph.
- line plot.
- frequency table.

It is my favorite because ... _____

_____.



Unit 12, Lesson 2: Creating Displays for Numerical Data Sets



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, and whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can distinguish between univariate and bivariate data, including which representations can be used to display each type of data set.
- ☐ I can create a graphical display for univariate numerical data.
- ☐ I can create a graphical display for bivariate numerical data.



12.2.1 Warm-Up: Classifying Data

When determining the best display for a data set, the first step is to determine the type of data you are being given.

Data can be classified as categorical or numerical, and univariate (one variable) or bivariate (two variables).

1. Identify the type of data for each example in the table.
The Pew Research Center collected data from a sample of 743 teens to analyze trends in social media and technology use.

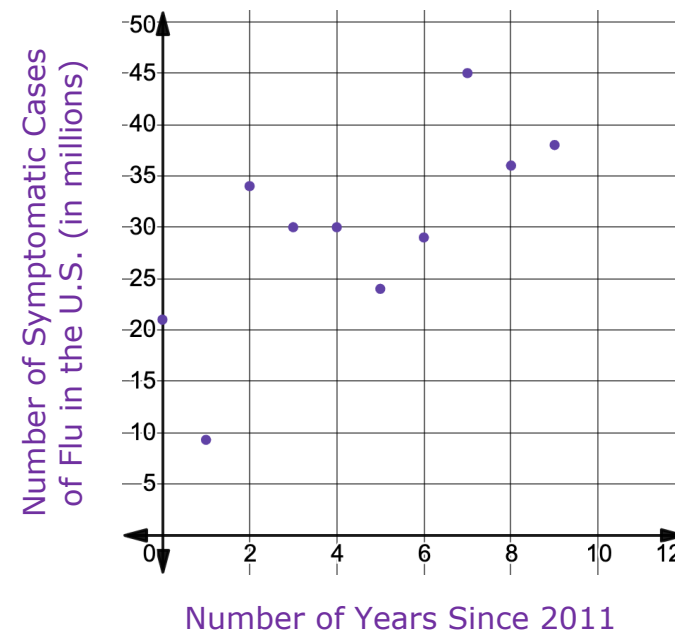
Example	Categorical or Numerical?	Univariate or Bivariate?
a. The number of people you consider close friends	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate
b. The number of people you consider close friends and the time spent on social media per day	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate
c. Your most used social media site (Twitter, Instagram, Facebook, Snapchat, YouTube, or Reddit)	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate
d. Your most used social media site (Twitter, Instagram, Facebook, Snapchat, YouTube, or Reddit) and what effect you would say social media has had on people your age (mostly positive, mostly negative, neutral)	☐ Categorical ☐ Numerical	☐ Univariate ☐ Bivariate



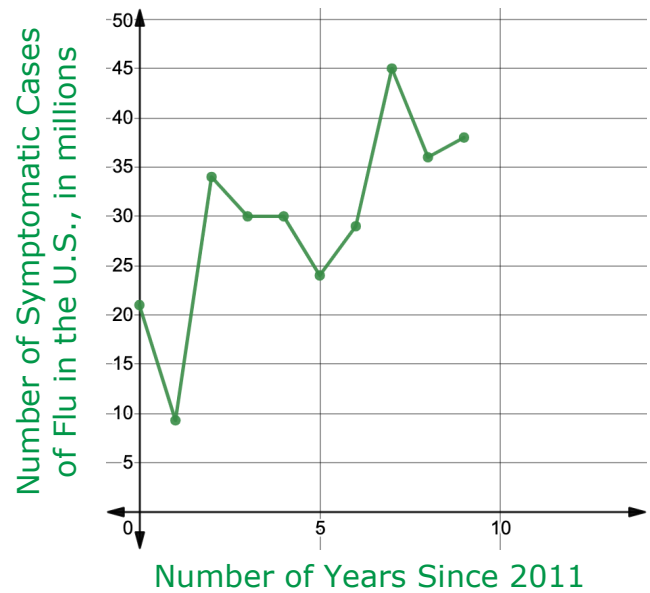
12.2.2 Refresher: Displays for Bivariate Numerical Data

Bivariate numerical data can be represented in different ways, including **scatter plots** and **line graphs**.

A scatter plot of the data on symptomatic cases of the flu in the U.S. is shown, where y is the number of symptomatic cases of the flu in the U.S., in millions, and x is the number of years since 2011.



The same data can be represented using a **line graph**.



1. Discuss with your partner the similarities and differences between the scatter plot and the line graph.

Complete the table.

Similarities Between Scatter Plots and Line Graphs	Differences Between Scatter Plots and Line Graphs

2. If data was collected on the number of symptomatic flu cases for each state for every year since 2011, which display would you choose to represent the data? Explain your reasoning.
3. What do you notice about the relationship between the number of symptomatic flu cases and years since 2011?



Both scatter plots and line graphs can be used to observe the relationship between two variables in a set of bivariate data. While overall trends can be seen in both graphs, line graphs allow for closer analysis from one point to another in the data set. Scatter plots are useful for larger sets of data.



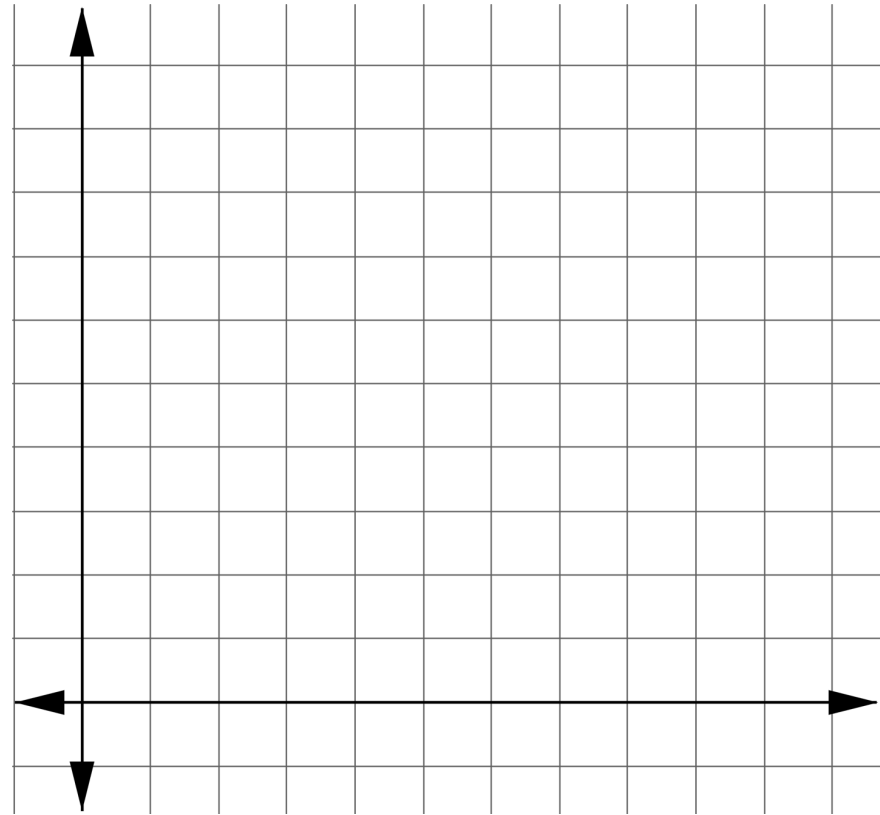
12.2.3 Try It: Creating Displays for Numerical Data

1. The prices and size, in carats, of 20 different diamonds are shown in the table.

Weight (carats)	Actual Price (\$)
1	3,772
1	4,221
1	4,032
1	5,385
1.05	3,942
1.05	4,480
1.06	4,511
1.2	5,544
1.3	6,131
1.32	5,872
1.41	7,122
1.5	7,474
1.5	5,904
1.59	8,706
1.61	8,252
1.73	9,530
1.77	9,374
1.85	8,169
1.9	9,541
2.04	9,125

- a. The data shown in the table is ☐ numerical
and ☐ univariate.
☐ bivariate.
- b. The data shown in the table should be displayed in
a ☐ scatter plot because ...
☐ line graph

- c. Create a display to represent the relationship between size and price of a diamond.



2. Laverne and Rocco planned to travel the greatest distance between any two points in the United States — from Log Point, Elliot Key, Florida to Kure Atoll, Hawaii — a distance of 5,859 miles.

They took a ferry from Log Point to Miami, then drove to Los Angeles, California, where they took a plane to Hawaii. They drove anywhere from 5-8 hours a day, and kept track of how many miles they traveled per day on their trip. This information is shown below.

{224, 294, 309, 317, 321, 339, 355, 361, 383, 456, 2500}

- a. Complete the statement.

The data set is

☐ numerical
☐ categorical

 and

☐ univariate.
☐ bivariate.

- b. Select all of the displays that **can** be used to display the data.

- ☐ Dot plot
- ☐ Histogram
- ☐ Box plot
- ☐ Stem-and-leaf
- ☐ Bar graph
- ☐ Circle graph
- ☐ Scatter plot

- c. Select all of the displays that **are best** to display the data.

- ☐ Dot plot
- ☐ Histogram
- ☐ Box plot
- ☐ Stem-and-leaf
- ☐ Bar graph
- ☐ Circle graph
- ☐ Scatter plot

- d. Create an appropriate graphical representation for the data.

- e. If the value 2500 is removed from the data set, which display would you create to represent the data? Explain your reasoning.

3. Basil recorded the weights, rounded to the nearest pound, of 15 cats at a pet show.

4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6

- a. Create an appropriate graphical display for the data. Do not use the same type of display you created in the previous problem.
- b. What other displays could be used to display his data?



12.2.4 Bring It Together: Choosing the Best Graphical Representation for Numerical Data

Univariate, numerical data:

Line Plots (Dot Plots)



- Best for small to moderate sets of data
- Shows the shape and distribution of data
- Shows individual data values



Histograms

- Best for larger sets of data and data over a large range,
- Can be used for continuous data
- Individual data values are not shown, only frequency over intervals
- Shows shape and distribution of the data



Stem-and-Leaf Plots

- Best for small data sets
- Shows the shape and distribution of data
- Displays individual data values in a more compressed way than a dot plot



Box Plots

- Best for large sets of data
- Displays variation of the data
- Individual data values are not shown
- Shows the 5-number summary of the data set

Numerical, bivariate data:



Scatter plots

- Best for large sets of data
- Helpful for looking at trends over a large data set
- Useful for comparing two quantities without thinking of either one as the independent or dependent variable



Line Graphs

- Best for smaller sets of data
- Helpful for looking at small changes over time



12.2.5 Wrap-Up: Reflection

Complete each statement based on today's lesson.

1. Today I am still unsure about ...

2. One question I still have is ...